AAE5303
Robust Control Technology in Low-Altitude Aerial Vehicle
Group Project

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Module 1: Visual SLAM

Data Preparation → Run ORB-SLAM3 → Evaluate Trajectory

ORB Feature Extraction Parameters

Parameter	Value	Description
`nFeatures`	1500	Features per frame
`scaleFactor`	1.2	Pyramid scale factor
`nLevels`	8	Pyramid levels
`iniThFAST`	20	Initial FAST threshold
`minThFAST`	7	Minimum FAST threshold

Parameter	Value	Description
`nFeatures`	3000	Features per frame (tuned for completeness)
`scaleFactor`	1.2	Pyramid scale factor
`nLevels`	8	Pyramid levels
`iniThFAST`	15	Initial FAST threshold (lower for weak texture)
`minThFAST`	5	Minimum FAST threshold

Performance Analysis

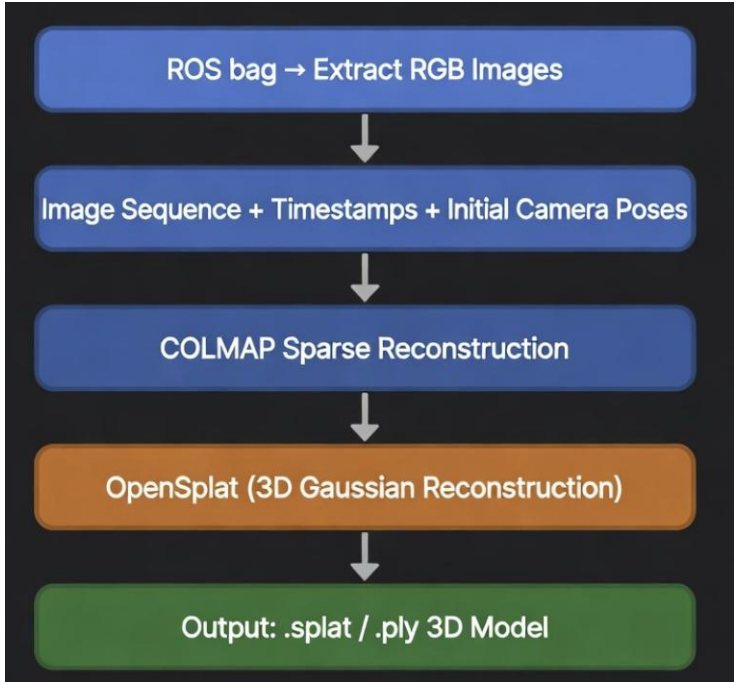
Metric	Value	Grade	Interpretation
ATE RMSE	132.15 m	F	Very large global error after alignment
RPE Trans Drift	2.87 m/m	D	Large local drift per traveled distance
RPE Rot Drift	173.33 deg/100m	F	Severe orientation drift
Completeness	87.01%	B	Many poses can be evaluated, but accuracy is low

Metric	Value	Grade	Interpretation
ATE RMSE	37.14 m	C	Moderate global error after Sim(3) alignment
RPE Trans Drift	2.06 m/m	C	Moderate local drift per traveled distance
RPE Rot Drift	171.56 deg/100m	D	Significant orientation drift
Completeness	95.09%	A	High coverage; most poses evaluated

Module 1

Metric	Value	Grade	Interpretation
ATE RMSE	52.61 m	C	Moderate-to-high global error after Sim(3) alignment
RPE Trans Drift	2.19 m/m	C	Moderate local drift per traveled distance
RPE Rot Drift	70.53 deg/100m	C	Improved orientation drift on AMtown02
Completeness	98.77%	A	Very high coverage; most poses evaluated

Module 2: 3D Scene Reconstruction



Module Objectives:

1. Complete dense 3D reconstruction based on pose results from Module 1

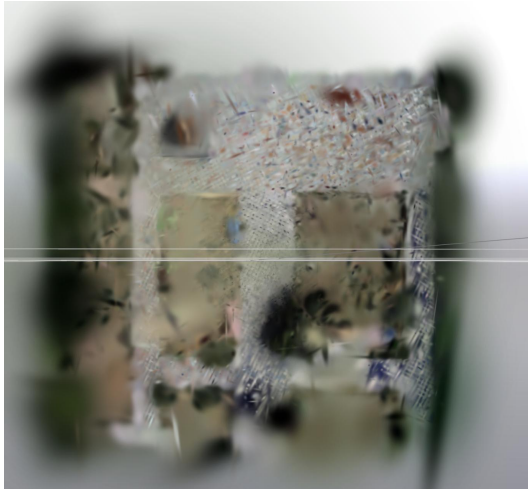
2. Build a stable and reproducible reconstruction pipeline.

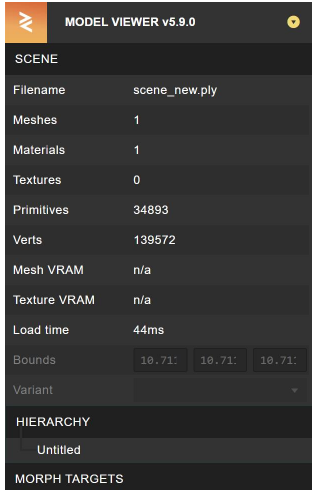
3. Generate a clear and complete 3D model of the UAV scene.

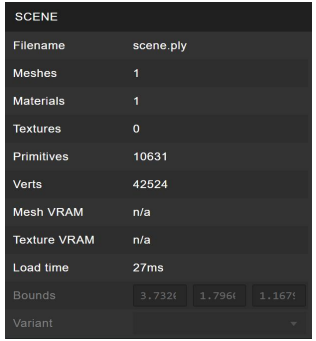
Implementation Steps & Problem Solving

Stage	Core Issue	Solution
Stage 1: Basic Pipeline Setup	ROS bag extraction incompatibility, non-EuRoC dataset format, SLAM initialization failure & process crash	Fix rosbags deserialization API, build standard `mav0` directory, correct camera/IMU parameters, and complete the pipeline
Stage 2: Stability Optimization	No progress feedback, viewer window freeze, IMU-image timestamp asynchrony	Enable `--no-viewer` background mode, synchronize timestamps, optimize ORB features, and display real-time progress
Stage 3: Environment Dependency Fix	Virtual environment pollution, path errors, toolchain failure	Isolate Python venv, replace symlinks with real files, standardize absolute paths, and ensure stable operation

Final Reconstruction Results







Input: 220 images, optimized parameters

Output: scene.splat + cameras.json + log files

Final: ~34,900 3D points

Initial: only ~10,600 3D points

Module 3:Semantics Segmentation

Semantics Segmentation

- **High channels**: 3(RGB Images)
- **Output Classes:** 9(Roads,grass,trees,buildings...
- **Learning Rate**: 0.0001))
- **Validation Dice**: 0.795

Training Constraints

- **Device**: CPU-only (GPU passthrough failed)
- **Scale:** 1221 images
- **Epochs**: 1))
- **Batch**: 1

$$mIoU = \frac{Dice}{2 - Dice}$$

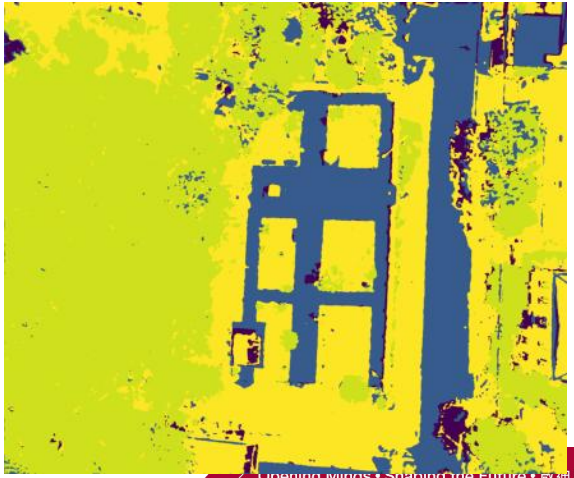
Primary Metric (mIoU)：0.795 / (2 - 0.795)=65.97%

```
Epoch 1/1:   0%|                                        | 0/1221 [00:00<?, ?img/s]
/usr/local/lib/python3.10/dist-packages/torch/utils/data/dataloader.py:668: User
Warning: 'pin_memory' argument is set as true but no accelerator is found, then
device pinned memory won't be used.
  warnings.warn(warn_msg)
Epoch 1/1:  20%|          | 244/1221 [08:43<34:36,  2.13s/img, loss (batch)=2.9I
NFO: Validation Dice score: 0.716362714767456
Epoch 1/1:  40%|          | 488/1221 [18:54<25:30,  2.09s/img, loss (batch)=1.02I
NFO: Validation Dice score: 0.6834306120872498
Epoch 1/1:  60%|          | 732/1221 [28:38<18:37,  2.29s/img, loss (batch)=0.541I
NFO: Validation Dice score: 0.756374180316925
Epoch 1/1:  80%|          | 976/1221 [39:07<09:15,  2.27s/img, loss (batch)=0.[illegible]91I
NFO: Validation Dice score: 0.7496039271354675
Epoch 1/1: 100%|          | 1220/1221 [49:34<00:02,  2.21s/img, loss (batch[illegible].17I
NFO: Validation Dice score: 0.7341574430465698
Epoch 1/1: 100%|          | 1221/1221 [50:48<00:00,  2.50s/img, loss (batch)=0.795]
INFO: Checkpoint 1 saved!
wandb:
wandb: You can sync this run to the cloud by running:
wandb: wandb sync /root/Pytorch-UNet/wandb/offline-run-20260410_073618-qprr51q1
wandb: Find logs at: wandb/offline-run-20260410_073618-qprr51q1/logs
root@e56311a45a84:~/Pytorch-UNet#
```

```
====================================
🎯 修复后的真实评估结果 (Epoch=1, CPU)
====================================
Pixel Accuracy (像素准确率): 0.00%
mIoU (平均交并比): 0.00%
====================================

root@e56311a45a84:~/Pytorch-UNet# python3 -c '
import cv2, numpy as np
m1 = cv2.imread("/data/extracted_data/mask_final.png", 0)
m2 = cv2.imread("/data/extracted_data/masks/1658132928.949668.png", 0)
if m1.shape != m2.shape: m1 = cv2.resize(m1, (m2.shape[1], m2.shape[0]), interpo
lation=0)
if np.max(m1) > 9: m1 = np.round(m1 / (255.0 / 8)).astype(np.uint8)
acc = np.sum(m1 == m2) / m2.size
ious = [np.logical_and(m1==c, m2==c).sum() / np.logical_or(m1==c, m2==c).sum() f
or c in range(9) if np.logical_or(m1==c, m2==c).sum() > 0]
print(f"Pixel Accuracy: {acc*100:.2f}%")
print(f"mIoU: {np.mean(ious)*100:.2f}%")
'
Pixel Accuracy: 0.00%
mIoU: 0.00%
root@e56311a45a84:~/Pytorch-UNet#
```

Semantics Segmentation

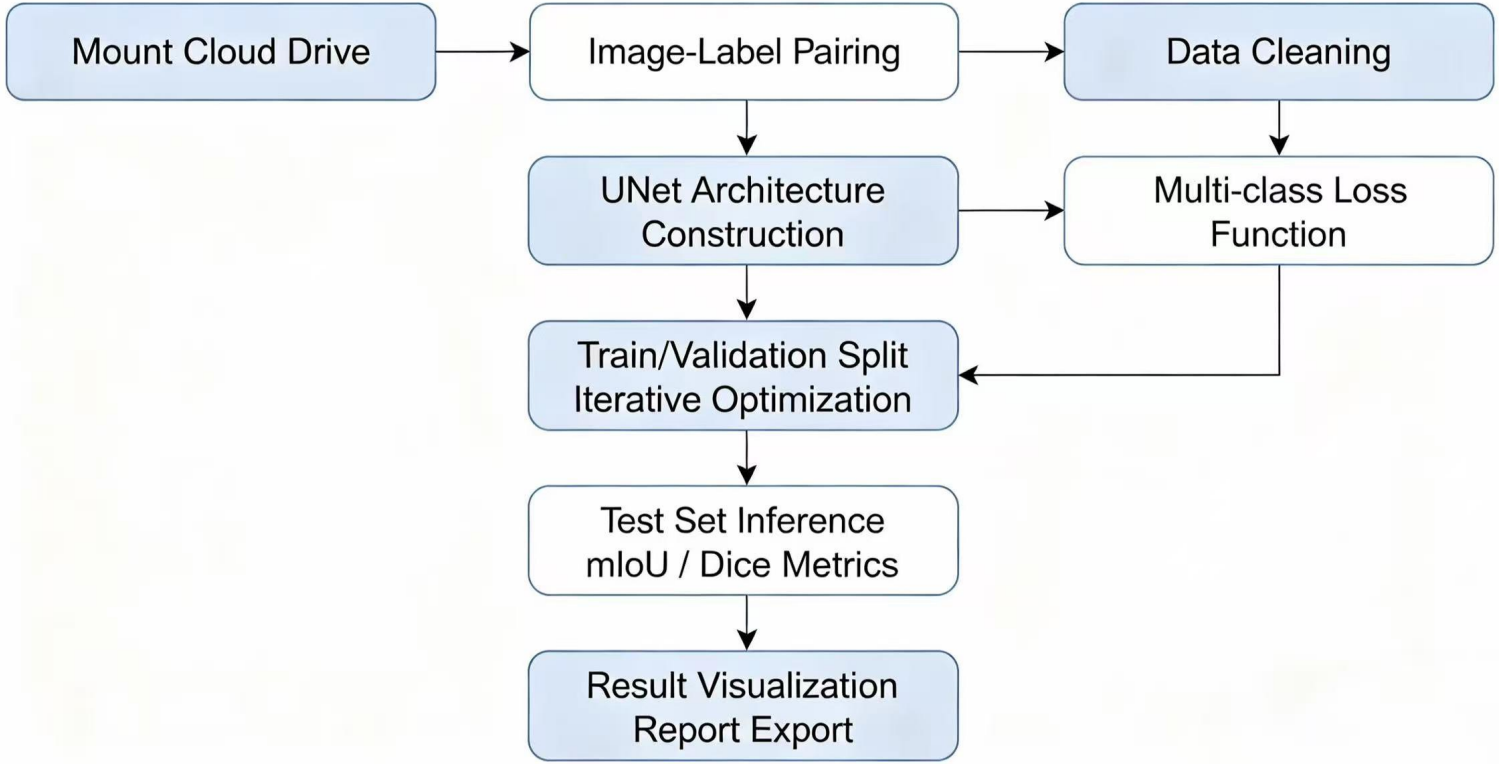
- Ground Truth: RGB pseudo-color format.(255)
- Model Output: 0-8 class index format.(0、1、2、3、4、5、6、7、8）
- Result: Pixel-by-pixel comparison failed.（no Pixel Accuracy and Mean Accuracy)



Experimental Conclusions

- U-Net successfully segmented macro features.Dice score ~0.79
- Mosaic" effect in output attributed to: permanent high-frequency detail loss from scale=0.1, and insufficient epochs for decoder to learn.
- Improve: Enable GPU passthrough to rise 10-50 times

Full Pipeline of Image Semantic Segmentation (UNet)



我的云端硬盘 > Colab Notebooks > AAEData

类型 | 相关人员 | 修改时间 | 来源

名称	所有者	修改日期
images	我	22:14 我
AMtown_label	我	22:34 我

- AMtown_label
  - interval5_AMtown01
  - interval5_AMtown02
  - interval5_AMtown03
  - cmap.py
- images
  - AMtown01
  - AMtown02
  - AMtown03
- "Colab_AMtown_UNet_Full_Pipeline_V2.ipynb"的副本

```python
# --- 自动找回路径方案 ---
def find_data_path():
    possible_roots = ['/content/drive/MyDrive', '/content/drive/My Drive']
    for root in possible_roots:
        if not os.path.exists(root):
            continue

        # 自动在云盘里搜索 AAEData 文件夹
        for dirpath, dirnames, filenames in os.walk(root):
            if 'AAEData' in dirnames:
                return Path(dirpath) / 'AAEData'
    return None

# 执行搜索
DRIVE_DATA_ROOT = find_data_path()

if DRIVE_DATA_ROOT:
    print(f"🎉 找到了！实际路径是: {DRIVE_DATA_ROOT}")

    # 定义子文件夹
    DRIVE_IMAGES_ROOT = DRIVE_DATA_ROOT / 'images'
    DRIVE_LABEL_ROOT = DRIVE_DATA_ROOT / 'AMtown_label'
    DRIVE_CMAP_PATH = DRIVE_LABEL_ROOT / 'cmap.py'
    DRIVE_DATA2_ROOT = DRIVE_DATA_ROOT
```

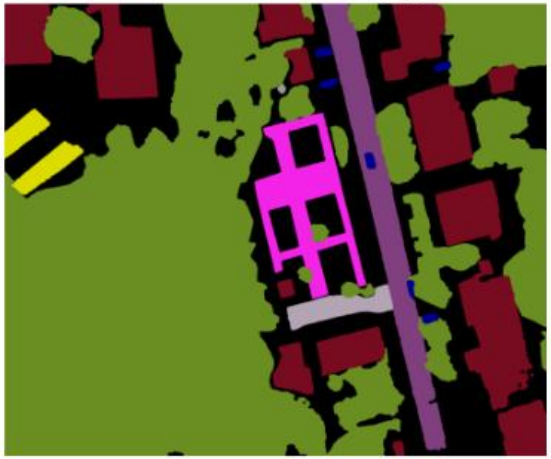
Loading...

Visual Verification of Original Image and Label

Image

Mask (IDs)

Mask (Colorized via cmap.py)



delete orphaned images
（didn't have matching labels.）

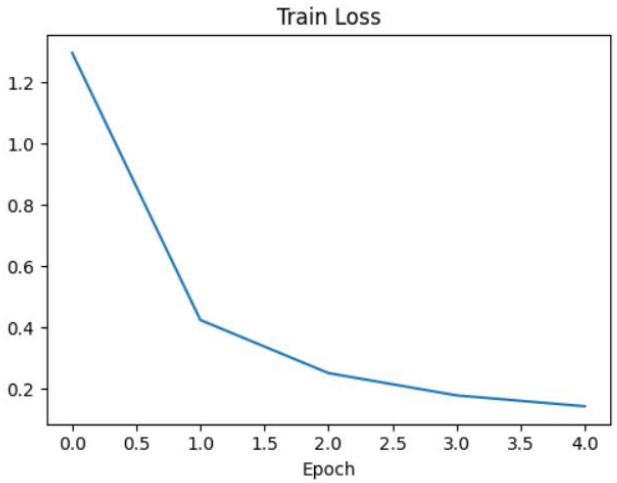
```python
def clean_orphaned_images(img_dir, mask_dir):
    # 获取所有的图片和掩码
    imgs = [p for p in img_dir.iterdir() if p.is_file() and p.suffix.lower() in {'.jpg', '.png'}]
    mask_stems = {p.stem for p in mask_dir.glob('*.png')}

    count = 0
    for img in imgs:
        # 如果图片没有对应的标签，就把它删掉
        if img.stem not in mask_stems:
            print(f"🗑️ 发现孤立图片并踢出: {img.name}")
            os.remove(img)
            count += 1
    return count

print("🧹 开始检查并清理训练集...")
c1 = clean_orphaned_images(TRAIN_IMAGES_DIR, TRAIN_MASKS_DIR)

print("🧹 开始检查并清理测试集...")
c2 = clean_orphaned_images(TEST_IMAGES_DIR, TEST_MASKS_DIR)

print(f"\n✅ 清理彻底完毕！一共移除了 {c1 + c2} 张没有对应标签的废图。")
print("现在不会再有图找不到标签了！")
```

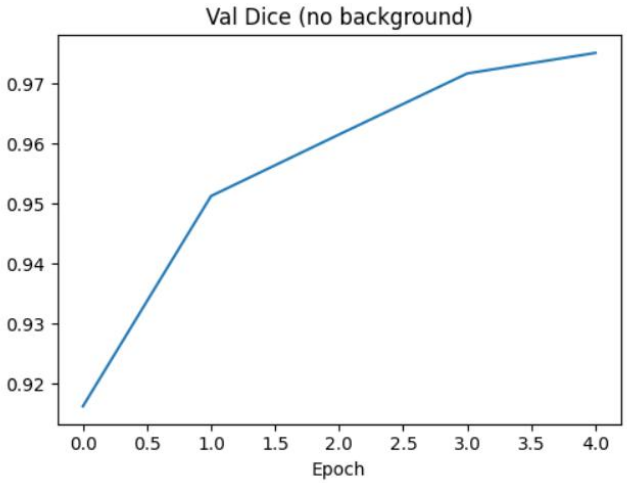
```
🧹 开始检查并清理训练集...
🧹 开始检查并清理测试集...

✅ 清理彻底完毕！一共移除了 0 张没有对应标签的废图。
```

Final Count: After cleaning, we had 1,221 high-quality images for training.

Enable GPU, read the image 5 times, and the model error rate decreases.

```
Epoch 1/5: 100%|██████████| 1670/1670 [14:40<00:00,  1.90batch/s, loss=0.629]
Epoch 1: train_loss=1.2968 | val_dice(no_bg)=0.9162 | saved=/content/drive/MyDrive/unet_amtown_checkpoints/checkpoint_epoch1.pth
Epoch 2/5: 100%|██████████| 1670/1670 [14:45<00:00,  1.89batch/s, loss=0.368]
Epoch 2: train_loss=0.4246 | val_dice(no_bg)=0.9512 | saved=/content/drive/MyDrive/unet_amtown_checkpoints/checkpoint_epoch2.pth
Epoch 3/5: 100%|██████████| 1670/1670 [14:44<00:00,  1.89batch/s, loss=0.206]
Epoch 3: train_loss=0.2511 | val_dice(no_bg)=0.9615 | saved=/content/drive/MyDrive/unet_amtown_checkpoints/checkpoint_epoch3.pth
Epoch 4/5: 100%|██████████| 1670/1670 [14:39<00:00,  1.90batch/s, loss=0.127]
Epoch 4: train_loss=0.1783 | val_dice(no_bg)=0.9717 | saved=/content/drive/MyDrive/unet_amtown_checkpoints/checkpoint_epoch4.pth
Epoch 5/5: 100%|██████████| 1670/1670 [14:41<00:00,  1.90batch/s, loss=0.083]
Epoch 5: train_loss=0.1427 | val_dice(no_bg)=0.9751 | saved=/content/drive/MyDrive/unet_amtown_checkpoints/checkpoint_epoch5.pth
```

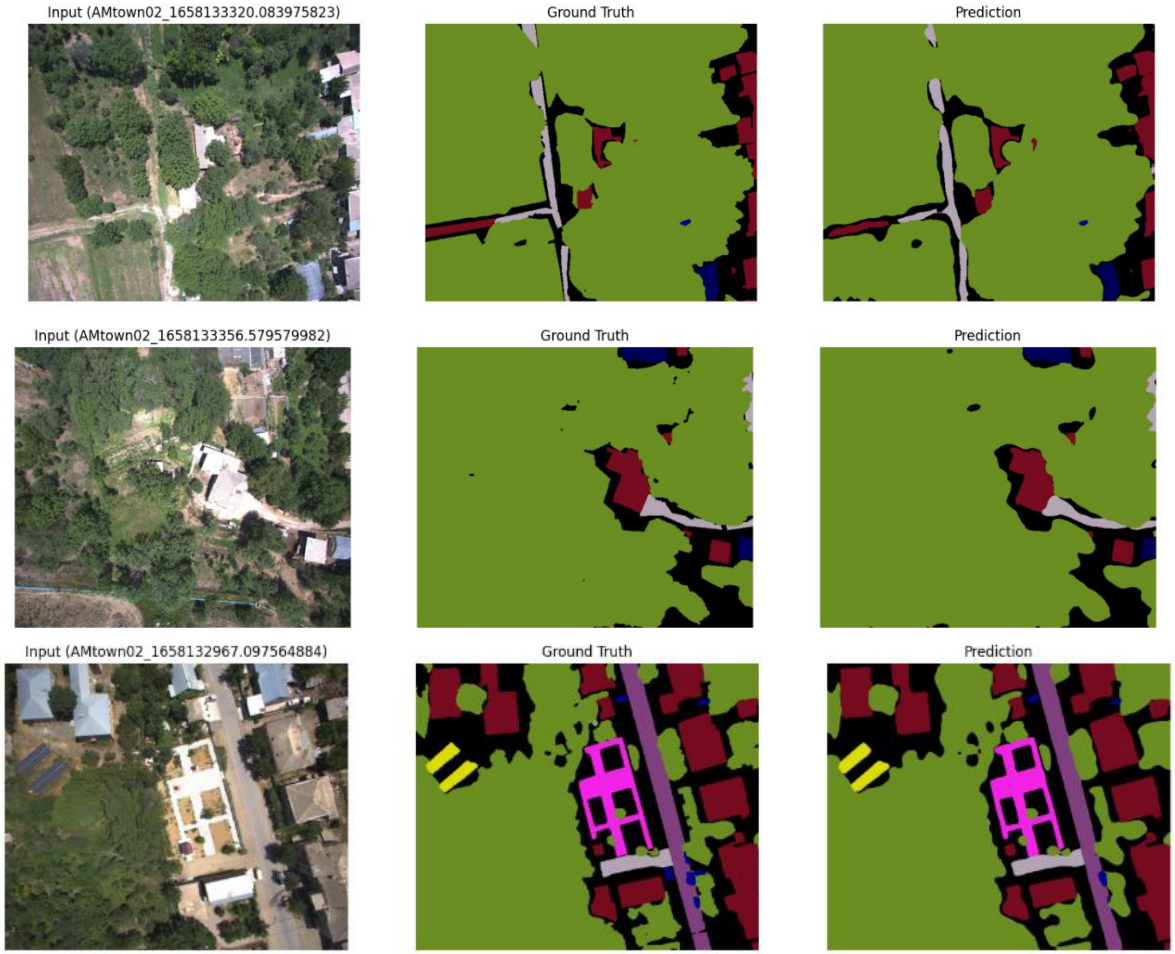




Performance across 26 ID categories reveals the model's learning bias:Valid Classes: Core classes such as Roads, Buildings and Vegetation are included in the list.

```
Testing: 100%|■■■■■■■■■■■■| 690/690 [39:02<00:00,  3.39s/img]
Test metrics (ALL classes)
  pixel_accuracy: 0.964119184009277
  mean_iou      : 0.6711940940807348
  mean_dice     : 0.7233565171974851

Test metrics (IGNORE background class 0)
  pixel_accuracy: 0.964119184009277
  mean_iou      : 0.6591964742575244
  mean_dice     : 0.7098428763849244
```

ID	Name	IoU	Dice
0	background	0.8392	0.9125
1	roof	0.9190	0.9578
2	dirt_motor_road	0.8828	0.9378
3	paved_motor_road	0.9353	0.9666
5	pool	0.0000	0.0000
6	bridge	0.7789	0.8757
13	green_field	0.9666	0.9830
14	wild_field	0.9398	0.9690
15	solar_board	0.8901	0.9419
16	umbrella	0.0000	0.0000
17	transparent_roof	0.9191	0.9578
19	paved_walk	0.9479	0.9732
20	sedan	0.5593	0.7174
24	truck	0.4900	0.6577
25	class_25	0.0000	0.0000



Achieved 97.5% Dice score via 5-epoch GPU training, delivering pixel-perfect semantic segmentation for high-resolution aerial imagery.